GROUP 7
BSIT-S-4A

IT PROFESSIONAL 04

Cloud Service Models

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Overview of Cloud Service Models

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What is a Cloud Service Model?

It describes how cloud computing resources are packaged and delivered to users, defining the division of responsibility between the cloud provider and the customer.

THREE MAIN MODELS: IaaS, PaaS, SaaS

IaaS (Infrastructure as a Service)

Provider gives you basic building blocks. You handle everything else.

PaaS (Platform as a Service)

Provider manages the infrastructure and the platform. You just build and deploy your application.

SaaS (Software as a Service)

Provider manages everything. You just use the finished application.

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Infrastructure as a Service (IaaS)

Definition

It is a cloud computing model in which a cloud provider delivers fundamental IT infrastructure over the internet. Instead of buying and maintaining physical hardware, customers rent these resources on demand.

- The provider manages the physical hardware and data center.
- The customer manages the operating system, applications, and data.

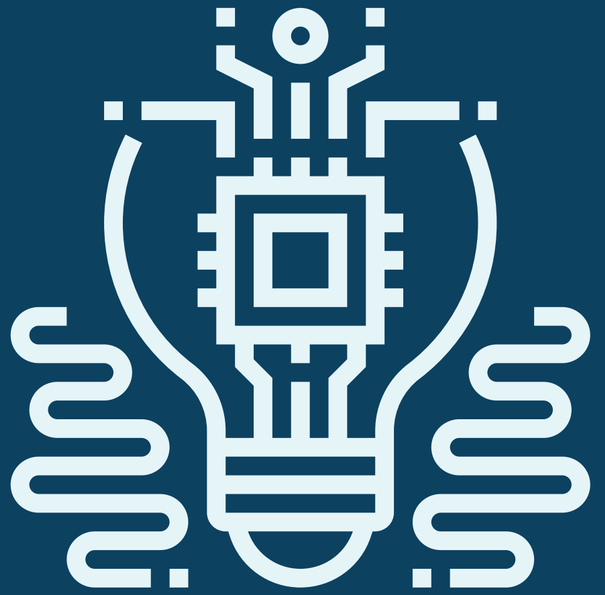
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Infrastructure as a Service (IaaS)

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Components

1. **Compute**
   - Provides processing power for running applications and workload.
   - Includes virtual machines (VMs), CPUs, and RAM.



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Infrastructure as a Service (IaaS)

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Components

2. Storage

- Provides space for storing files, databases, applications, and backups.
- Includes block storage, object storage, and virtual disks.



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Infrastructure as a Service (IaaS)

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Components

3. Networking

- Connects cloud resources and allows them to communicate with users and other systems.
- Includes virtual networks, IP addresses, routers, and firewalls.



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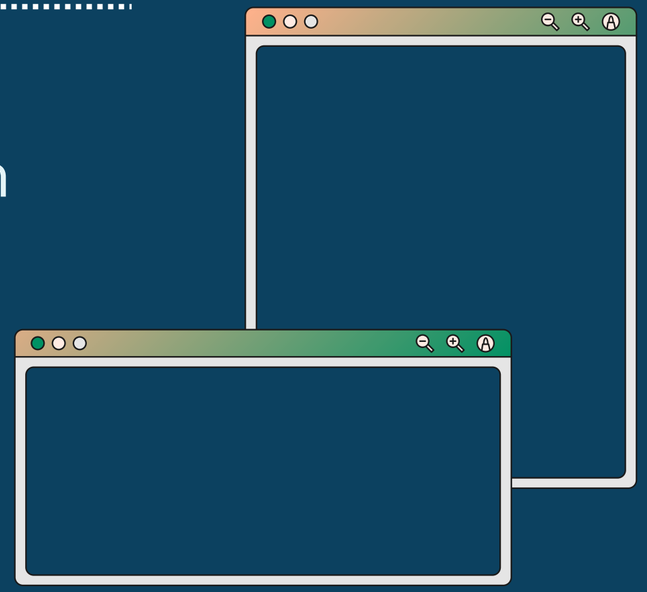
Infrastructure as a Service (IaaS)

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Components

4. Virtualization

- Allows multiple virtual machines to run on a single physical server, making hardware resources more efficient.



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Infrastructure as a Service (IaaS)

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Components

5. Security

- Protects cloud infrastructure, data, and applications.
- Includes firewalls, access controls, encryption, and identity management.



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PaaS (Platform as a Service)

Definition

It describes how cloud computing resources are packaged and delivered to users, defining the division of responsibility between the cloud provider and the customer. The PaaS provider hosts everything—servers, networks, storage, operating system software, databases and development tools—at their data center.

Benefits

- ***Faster time to market***
- ***Affordable access to a wider variety of resources***
- ***More freedom to experiment, with less risk***
- ***Easy, cost-effective scalability***
- ***Greater flexibility for development teams***
- ***Lower overall costs***
- ***Enhanced security***

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PaaS (Platform as a Service)

Three Main Components:

- **Cloud infrastructure**, including virtual machines (VMs), operating system software, storage, networking and cloud security features, such as encryption and firewalls.
- **Software for building**, deploying and managing applications.
- **Graphic user interface (GUI)** where development or DevOps teams can do all their work throughout the entire application lifecycle.

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SaaS (Software as a Service)

Definition

Software as a service (SaaS) is application software hosted on the cloud and used over an internet connection through a web browser, mobile app or thin client. SaaS enables your organization to use an application without the expense of setting up the infrastructure to run it and the effort and personnel to maintain it.

SaaS uses the Internet to deliver subscription software services that are managed by third-party vendors. Well-known SaaS examples include **Dropbox**, **Google Workspace, and Salesforce.**

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SaaS (Software as a Service)

The Main Components:

- **Infrastructure** – servers, storage, networking (hosting)
- **Database** – stores user/tenant data
- **Security** – authentication, authorization, encryption
- **Subscription/Billing Management** – handles payments and licensing

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SaaS (Software as a Service)

Advantages

- *Accessible from anywhere*
- *Cost-effective*
- *Easy to implement, update, and debug*
- *Easy to scale*

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SaaS (Software as a Service)

Disadvantages

- *Increased security risks*
- *Slower speed*
- *Loss of control*
- *Lack of customization*

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Comparison of IaaS, PaaS and SaaS

Comparison by PURPOSE

- ***IaaS:*** *Provides infrastructure for running systems.*
- ***PaaS:*** *Provides an environment for developing and deploying applications.*
- ***SaaS:*** *Provides finished applications for users.*

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Comparison of IaaS, PaaS and SaaS

Comparison by CUSTOMIZATION

- ***IaaS:*** *You can customize the operating system, networking, storage, and deployed applications.*
- ***PaaS:*** *You customize your application, but the underlying platform is controlled by the provider.*
- ***SaaS:*** *You generally customize only the settings and features offered by the application.*

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Comparison of IaaS, PaaS and SaaS

Comparison by TECHNICAL SKILL REQUIRED

- ***IaaS:*** *Requires knowledge of servers, operating systems, networking, and security.*
- ***PaaS:*** *Primarily requires programming and application-development knowledge.*
- ***SaaS:*** *Usually requires only knowledge of how to use the application.*

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Comparison of IaaS, PaaS and SaaS

Layer	Who manages Infrastructure	Who manages Os/ Runtime	Who manages Apps/ Data	User Controls	Level of control	typical users	Example
IaaS	Provider	Consumer	Consumer	OS, storage, deployed apps, networking	HIghest	System admin	AWS EC2, Microsoft Azure VMs
Paas	Provider	Provider	Consumer	Applications and data only	Medium	Developers	Google App Engine, Heroku
Saas	Provider	Provider	Provider	Just uses the application	Lowest	End Users	Gmail, Salesforce, Office 365

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Everything as a Service (XaaS)

XaaS stands for Everything as a Service or Anything as a Service. It is a big group name for any tool, tech, or program sent to you over the internet. Instead of buying physical parts or programs for your office, you rent them on the cloud and pay as you go.

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Examples

FaaS (Function As A Service)

Run code in small functions, no server management (serverless)

Ex: AWS Lambda

Daas (Desktop As A Service)

Virtual desktops hosted and delivered over the cloud

Ex: Amazon Workspaces

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Examples

DBaaS (Database As A Service)

Fully managed databases. No setup or maintenance

Ex: Amazon RDS, Firebase

Caas (Containers As A Service)

Platform to deploy and manage containers at scale

Ex: Google Kubernetes Engine

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Benefits of XaaS

- Lower upfront costs
- Scalability
- Faster deployment
- Access to specialized technologies
- Less infrastructure maintenance
- Pay-as-you-use possibilities
- Easier adoption of new technologies

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Challenges of XaaS

Security and privacy

- Data and services are handled through external providers.

Internet dependency

- Services generally depend on network connectivity.

Vendor lock-in

- Switching providers may become difficult.

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Challenges of XaaS

Service availability

- Provider outages can affect users.

Cost management

- Using many services can result in unexpected or increasing costs.

Integration

- Different XaaS services may need to communicate with each other.


THANK YOU FOR LISTENING